ECE 329 Introduction to Electromagnetic Fields Spring 18

University of Illinois He, Goddard, Ilie

Exam 2

Wednesday, March 14, 2018 — 7:15-8:30 PM

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the boxes below.

Name:	Solutions
Section:	12 Noon 1 PM 2 PM

If you want your exam returned to a **different** section, write it here:

Special Return to Section:	12 Noon 1 PM 2 PM

This is a closed book exam and calculators/electronic devices are not allowed. You are allowed TWO double sided 3"x5" index card of notes. Please show all your work and make sure to include your reasoning for each answer. All answers should include **units** wherever appropriate. The exam is double sided. You may use the back of the exam as scratch paper.

Problem 1 (21 points)	
Problem 2 (20 points)	
Problem 3 (25 points)	
Problem 4 (14 points)	
Problem 5 (20 points)	
TOTAL (100 points)	

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1. (21 points) Multiple choice questions (3 points each). Circle only **ONE** letter next to the **ONE** correct statement. Note that most of these questions require some calculation to get the right answer, but you do not need to show your work to get credit. No partial credit is given on the multiple choice questions.

(i) Two parallel wires lie in the xy plane and carry currents $I_1 > 0$ and $I_2 > 0$, pointing in the same direction. The wires:

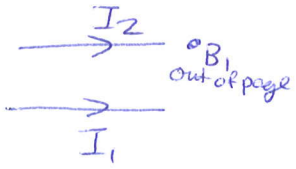
(a) experience an attractive force

(b) experience a repulsive force

(c) experience a force in the z direction

(d) experience no force between them

(e) None of these



$\vec{F}_2 = I_2 d\vec{l}_2 \times \vec{B}_1$ is towards the other wire

(ii) The magnetic flux density increases at a rate of $\frac{\partial \mathbf{B}}{\partial t} = 10\hat{z}\ [\frac{Wb}{m^2 \cdot s}]$. A 10 cm by 10 cm square conducting loop, centered at the origin in the x-y plane has 10 Ω of distributed resistance. The **magnitude** of the induced current in the conducting loop is:

(a) 1 mA

(b) 10 mA

(c) 0.1 mA

(d) 100 mA

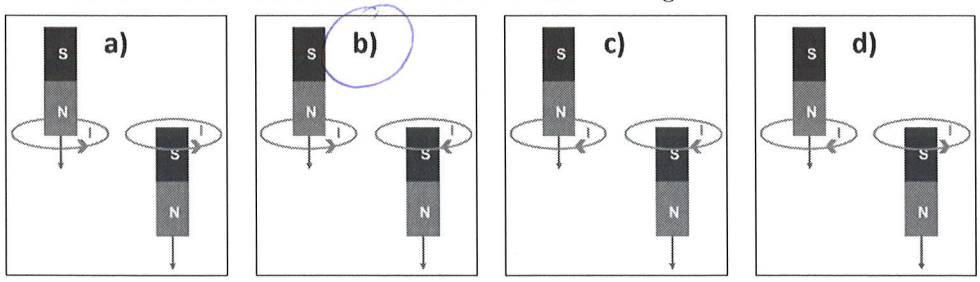
(e) None of these

$I = \left|\frac{\mathcal{E}mf}{R}\right| = \frac{\left|-\frac{d\Phi_B}{dt}\right|}{10}$

$\frac{d\Phi_B}{dt} = 10 \cdot 0.1 \cdot 0.1 = 0.1\ \frac{Wb}{s}$

$\Rightarrow I = 0.01\ A = 10\ mA$

(iii) A bar magnet is dropped through a conductive ring. Circle the one correct picture that shows the direction of the induced current in the conductive ring:



↓dB so $I_{induced}$ gives a B upward

↑dB so $I_{induced}$ gives a B downward

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(iv) If the current density $\mathbf{J} = 0$ within a certain volume in free space, then in this volume which of the following statement must be true for electric displacement field $\mathbf{D}$?

a) $\frac{\partial \mathbf{D}}{\partial t} = 0$

b) $\frac{\partial(\nabla \cdot \mathbf{D})}{\partial t} = 0$

c) $\nabla \cdot \mathbf{D} = 0$

d) $\mathbf{D} = 0$

e) None of the above

Continuity equation: $\nabla \cdot J = -\frac{\partial \rho}{\partial t}$

$\nabla \cdot D = \rho$ so $0 = \nabla \cdot J = -\frac{\partial}{\partial t}(\nabla \cdot D)$

(v) If the scalar and vector potential are given by $\Phi = 3x^2$ $[V]$ and $\mathbf{A} = -6xt\hat{x} + 3y\hat{y}$ $[\frac{Vs}{m}]$, respectively, then the electric field and the magnetic flux density satisfy

a) Both $\mathbf{E}$ and $\mathbf{B}$ are zero everywhere for all of time

b) $\mathbf{E}$ is constant in time but varies with position while $\mathbf{B}$ is constant in position but varies with time

c) $\mathbf{E}$ is constant in time but varies with position while $\mathbf{B}$ is constant in time but varies with position

d) $\mathbf{E}$ is constant in position but varies with time while $\mathbf{B}$ is constant in position but varies with time

e) $\mathbf{E}$ is constant in position but varies with time while $\mathbf{B}$ is constant in time but varies with position

f) None of the above

Note $\nabla \times A = 0$ so $B = 0$

$E = -\nabla\Phi - \frac{\partial A}{\partial t} = -6x\hat{x} + 6x\hat{x} = 0$

This is an interesting result. We have a complicated Φ and $\vec{A}$ but their associated fields $\vec{E} = \vec{B} = 0$ can also be described by $\Phi = 0$, $\vec{A} = 0$. See gauge transformations. The potentials are only defined up to a scalar function $\lambda(\vec{r}, t)$

(vi) Consider a sheet current $\mathbf{J_s} = 2\hat{y}$ A/m at $z = 0$ plane surrounded by free space. The magnetic field $\mathbf{H}$ in region $z < 0$ is found to be $\mathbf{H} = 3\hat{z}$ A/m. The $\mathbf{H}$ in region $z > 0$ can be determined as

a) $\mathbf{H} = 1\hat{z}$ A/m

b) $\mathbf{H} = 5\hat{z}$ A/m

c) $\mathbf{H} = -2\hat{x} + 3\hat{z}$ A/m

d) $\mathbf{H} = 2\hat{x} + 3\hat{z}$ A/m

e) $\mathbf{H} = -2\hat{y} + 3\hat{z}$ A/m

f) $\mathbf{H} = -2\hat{y} + 3\hat{z}$ A/m

$\hat{a}_n$ 1

$J_s = 2\hat{y}$ at $z = 0$

$3\hat{z} = H$

2

$B_\perp$ is continuous $B_{1n} = B_{2n}$

$\mu_0 H_{1n} = \mu_0 H_{2n}$

$\Rightarrow H_{1n} = 3$

$(H_1 - H_2)_t = J_s \times \hat{a}_n$

$= 2\hat{y} \times \hat{z} = 2\hat{x}$

$\Rightarrow H_{1t} = 2\hat{x}$

(vii) A wave propagates in free space. An instrument observes the phase change of the wave to be 3π radians in a 10 μs interval. The wavelength of the wave is approximately

a) 0.1 m

b) 0.5 m

c) 2 m

d) 5 m

e) 2000 m

f) None of these

$\omega = \frac{3\pi}{10\mu s}$

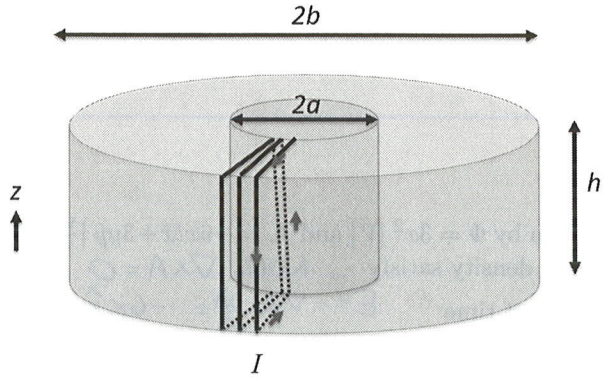
$v = \lambda f = c$

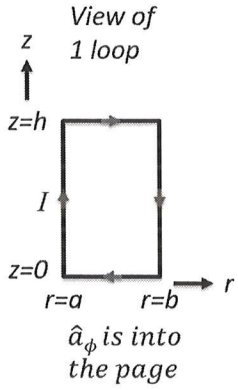
$\Rightarrow \lambda = \frac{c}{f} = \frac{c}{\omega} \cdot 2\pi = \frac{2\pi \cdot 3 \cdot 10^8 \cdot 10^{-5}}{3\pi}$

$= 2000\,m$

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2. (20 points) A coil is wound uniformly on a torus of of rectangular cross section, consisting of N turns of wire. The left picture below shows 3 closely spaced turns but note that these turns go all the way around the circumference of the torus. The right picture below shows the cross section for a single loop. The current flowing in the coil produces a magnetic flux density given by $\mathbf{B} = \frac{\mu_0 N I}{2\pi r}\hat{a}_\phi$, where r is the radial distance from the axis of the coil.





a) (10 pts) Find the magnetic flux through one turn of the coil

$$\Psi_1 = \iint B \cdot dS = \int_0^h dz \int_a^b dr \, \frac{\mu_0 N I}{2\pi r} = h \cdot \frac{\mu_0 N I}{2\pi} \ln\frac{b}{a}$$

Your Answer (include appropriate units): $\psi_1 =$ $\frac{\mu_0 N I h}{2\pi} \ln\frac{b}{a}$ [Wb]

b) (4 pts) Find the flux threading through the coil of N turns

N times as much

Your Answer (include appropriate units): $\psi_N =$ $\frac{\mu_0 N^2 I h}{2\pi} \ln\frac{b}{a}$ [Wb]

c) (6 pts) Find the **magnitude** and **direction** of the induced EMF if the current I is increasing in time

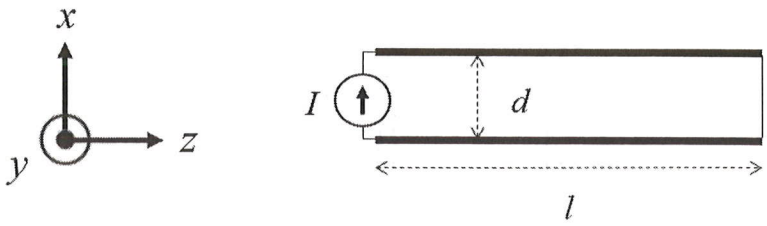
$$EMF = -\frac{d\Psi}{dt} = -\frac{\mu_0 N^2 h}{2\pi} \ln\frac{b}{a} \frac{dI}{dt}$$

Your Answer (include appropriate units): $EMF =$ $-\frac{\mu_0 N^2 h}{2\pi} \ln\frac{b}{a} \frac{dI}{dt}$ [V]
(circle one) clockwise or (counterclockwise) based on the above "View of 1 loop" drawing

because negative sign on Emf means it is going opposite the drawn curve

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3. (25 points) Consider in free space a parallel-plate transmission line composed by two thin conductor plates with plate width w, length l, and the plate separation $d << w$, l. The line is shorted at one end and a current I [A] flows along the top plate and back along the bottom plate. Assume the surface current density is uniform on both plates.



a) (4 pts) Find the surface current density vector $\boldsymbol{J_s}$ on the top and bottom plate.

Your Answer (include appropriate units and direction):
$\boldsymbol{J}_{s,top} = \frac{I}{w}\hat{z}$ [A/m] $\quad \boldsymbol{J}_{s,bottom} = -\frac{I}{w}\hat{z}$ [A/m]

b) (5 pts) Find the magnetic flux density vector $\boldsymbol{B}$ between the plates.



$B_{due\ to\ top} = \frac{1}{2}\mu_0 J_s \times a_n$ $\quad \hat{z}\times(-\hat{x}) = -\hat{y}$ $\quad$ Bottom plate adds B in between

Your Answer (include appropriate units): $\boldsymbol{B} =$ $\mu_0 \frac{I}{w}(-\hat{y})$ [Wb/m²]

c) (5 pts) Calculate the magnetic flux ψ linked by the current path (i.e. the flux through the loop of current).

$\psi = \iint_S B\cdot dS = B \cdot Area =$

Your Answer (include appropriate units): $\psi =$ $\mu_0 \frac{I}{w} \cdot ld$ [Wb]

d) (5 pts) Calculate the inductance L of the transmission line.

$L = \frac{\psi}{I}$

Your Answer (include appropriate units): $L =$ $\mu_0 \frac{ld}{w}$ [H]

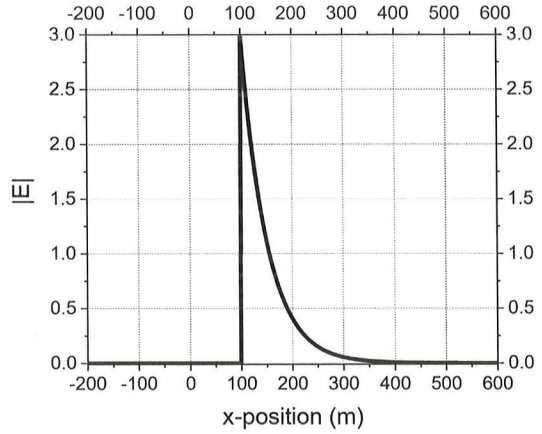
e) (6 pts) If instead the plate separation doubled to $2d$ and current reduced to $\frac{I}{2}$, how would the values of $|\boldsymbol{B}|$, ψ and L change?

$\lvert\boldsymbol{B}\rvert$ would scale by a factor of (circle one):	0.25	(0.5)	1.0	2.0	4.0
ψ would scale by a factor of (circle one):	0.25	0.5	(1.0)	2.0	4.0
L would scale by a factor of (circle one):	0.25	0.5	1.0	(2.0)	4.0

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4. (14 points) The electric field of a wave is observed to move with $\mathbf{v} = -c\hat{x} = -300\hat{x}\ [\frac{m}{\mu s}]$ and to have the following exponentially decaying profile as x increases at time $t = 1\mu s$:

$\mathbf{E}(x, t = 1\mu s) = 3u[x - 100]exp\left[-\frac{x-100}{50}\right]\hat{z}$ [V/m] where u is the unit step function.



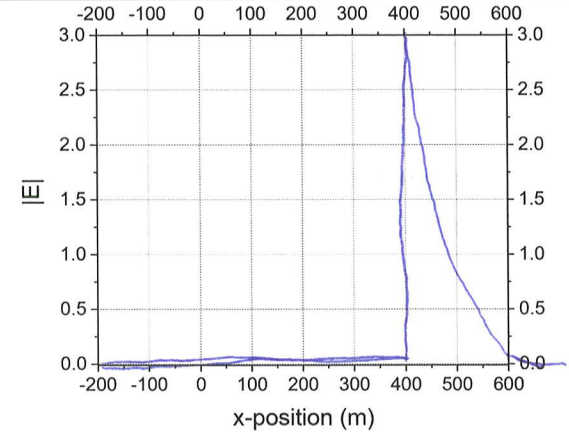
This wave is moving in $-\hat{x}$ direction meaning it was generated by a source near $x = +\infty$ at a time near $t = -\infty$ with a current profile that turns on (leading part of wave) and exponentially decays in time.

a) (7 pts) Find the expression for the electric field everywhere at $t = 0$ and also sketch the profile.

Wave moves left 300m per 1μs so 1μs ago, it was 300m shifted to the right.

Your Answer (include appropriate units and direction):

$\mathbf{E}(x, 0) =$ $3\ u(x-400)\ exp(-\frac{x-400}{50})\ \hat{z}$ [V/m]



b) (5 pts) Find the electric field $\mathbf{E}(x, t)$ for arbitrary x and t.

"Center" of wave, i.e. where it steps up from 0 to 3 is at $x = 400$ at $t = 0$ and moves left by 300 m/μs.

Your Answer (include appropriate units and direction):

$\mathbf{E}(x, t) = 3\ u(x - (400 + vt))\ exp\left(-\frac{(x-(400+vt))}{50}\right)\hat{z}$ [V/m] where $v = -300\frac{m}{\mu s}$

$= 3\ u(x - 400 + 300t)\ exp\left(-\frac{(x-400+300t)}{50}\right)\hat{z}$ [V/m] for x in meters and t in μs

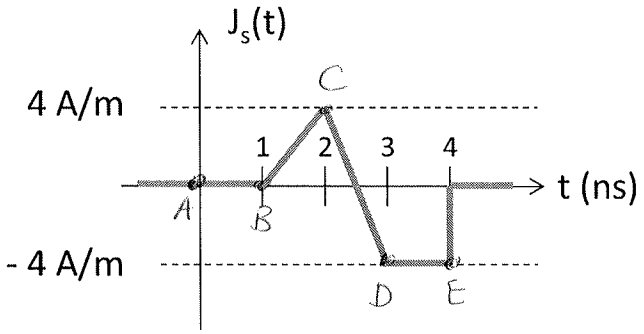
c) (2 pts) Circle one: TRUE or (FALSE) The wave is propagating in a conductive medium.

Propagation is at speed of light.

Also, the wave is not attenuating as it propagates. The amplitude is still 3 for the peak of the wave.

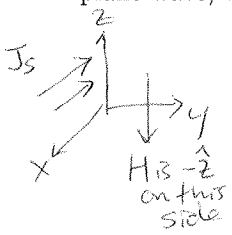
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5. (20 points) Consider an infinitesimally thin current sheet in the $y = 0$ plane that serves to launch forward and reverse uniform plane waves into free space. Let the expression for the current density be given by $\mathbf{J} = -J_s(t)\,\hat{x}$ [A/m] where $J_s(t)$ is shown below. Assume free space, $v_p = 3 \times 10^8$ m/s and $\eta = 120\pi$ [Ω].



a) (3 pts) Fill in the following expression for the magnetic field component of the **forward** uniform plane wave, which can be written in terms of $J_s(t)$ as:

H is +$\hat{z}$ on this side



$$\mathbf{H}(y,t) = \underline{\quad (ii) \quad} J_s(t - \frac{y}{v_p}) \underline{\quad (i) \quad} A/m$$

Your Answer:
(i) = unit vector signifying direction of **H** (circle one): $\hat{x}$ $\hat{y}$ ($\hat{z}$)

(ii) = coefficient in front of J_s (with proper polarity): $-1/2$

b) (3 pts) Fill in the following expression for the magnetic field component of the **reverse** uniform plane wave, which can be written in terms of $J_s(t)$ as:

$$\mathbf{H}(y,t) = \underline{\quad (ii) \quad} J_s(t + \frac{y}{v_p}) \underline{\quad (i) \quad} A/m$$

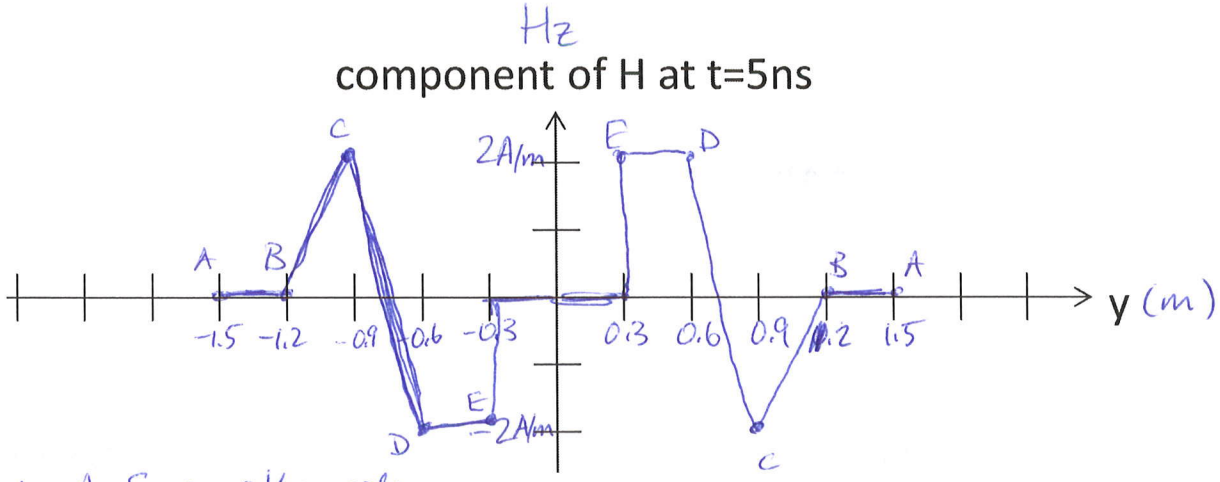
Your Answer:
(i) = unit vector signifying direction of **H** (circle one): $\hat{x}$ $\hat{y}$ ($\hat{z}$)

(ii) = coefficient in front of J_s (with proper polarity): $1/2$

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c) (14 pts) Sketch the appropriate component of **H** (i.e. the amount of H in the direction of $\hat{x}$, $\hat{y}$, or $\hat{z}$ depending on what your circled above) versus y at $t = 5$[ns]. Label the axis scales so that the magnitudes and positions can be easily read.

Hz



See points A-E on other side

Point A travels farthest: $d = vt = 3\cdot10^8$ m/s t = 300 m/μs · t = 0.3 m/ns · t

$d = 0.3$ m/ns · 5ns = 1.5m

Point B travels

$d = 0.3$ m/ns · 4ns = 1.2m

...

Point E travels

$d = 0.3$m

Pick horizontal axis as 0.3m per tick = 30cm

Amplitude is $\pm\frac{1}{2} J_s = \pm 2$ A/m

So each tick can be 1 A/m

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